MILITARY SPECIFICATION

PLOTTING BOARD: M17

1. SCOPE

1.1 This specification covers one type of plotting board known as Plotting Board: M17 and is used for plotting indirect fire data.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-F-14619	Fire Control Parts, Equipment, and Tools; Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U. S. Military Property

DRAWINGS

Ordnance Corps

F7689472	Plotting Board: M17

FSC 1220

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production one plotting board, representative of productions shall be made available to the Government to determine if the board meets all the requirements of this specification.

3.2 Materials.- The materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification

3.3 Construction.- The plotting board shall be manufactured in accordance with Drawing F7689472 "Plotting Board: M17", and all drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to insure the inherent quality and the performance capabilities of the product "prior to final acceptance by the Government.

3.4 General specification.- The contractor shall be responsible for adherence to and compliance with the following requirements of Specification MIL-F-13926:

a Order of precedence
b Protective finished surfaces
c Special accuracy
d Workmanship

3.5 Environmental.

3.5.1 Operating Temperature.- The plotting board shall meet the requirements of 3.6 through 3.9 thermally stabilized at ambient temperatures of minus 40 degrees and plus 125 degrees Fahrenheit.

3.5.2. Storage temperature.- The plotting board shall show no evidence of damage or defects; and the plotting board shall meet the requirements of 3.6 through 3.9 at room ambient temperature (plus 60 to 90 degrees Fahrenheit) after having been thermally stabilized at ambient temperatures of minus 60 degrees and plus 140 degrees Fahrenheit,

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3.6 Operability.- The dial shall rotate freely about its 6400 - mil gyration. The pivot pin shall not revolve in the base. The dial shall be removable from the base with normal finger pressure and without causing permanent distortion.

3.7 Marking and legibility.- The top face of the dial shall have a matte surface that will permit clear pencil markings and clean erasures. There shall be no visible defects such as pits, digs, scratches with depth, strains or other waves on the surface that may be mistaken for pencil markings, or shall cause lines or figures to appear distorted. Lines and figures on the base, as viewed through the dial shall be clear and legible without skips or brakes.

3.8 Accuracy.- With one extremity of the dial index line in coincidence with each of the diametrical center lines of the base. the other extremity of the index line shall be not more than 5 miles out of coincidence with the corresponding base center line.

3.8.1 The plotting board accuracy for the gun target (GT) line shall be within 5.0 mils as determined by placing pencil dots at the target (T) and gun (G) points as measured from observer (O) represented by the center pivoting point of the dial) using the OT and OG values specified below. When a line is drawn through the GT dots, the GT line is then layed parallel to the base "O" centerline and the GT mil value is determined from the mil scale on the dial (see 6.4).

	Problem 1		Problem 2		Problem 3	
	OT	OG	OT	OG	OT	OG
Mils	800	2500	5900	3850	2510	1310
Meters	2500	1250	3200	1050	1400	1800
	GT = 348 Mils		GT = 6159 Mils		GT = 2702 Mils (as read by the red numerals)	

3.9 Warpage.- There shall be no warpage in excess of 0.032 inch between the edge of the dial and the base.

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3.10 Identification marking - Unless otherwise specified, Identification marking shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's inspection.- Unless otherwise specified herein, the supplier is responsible for the performance of all Inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Production model.- Unless otherwise specified in the contract, one Plotting Board: M17 shall be selected by the Government Inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein,

4.2.1 Production model failure.- Should the production model fall to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When correction in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.3 Lot formation.- Inspection lot sizes and lot formation shall be inaccordance with Standard MIL-STD-105.

4.3.1 Defective units and lots shall be processed as specified in Standard MIZ-STD-105,

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4.5 Acceptance tests.

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T	e	x	t
T	e	x	t
T	e	x	t
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T	e	x	t
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T	e	x	t

MAJOR: AQL **1.0%**

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T	e	x	t	Text	Text	T e x t
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3.9 4.6.1 Standard Meas. Equipment (SME)

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TABLE I - CLASSIFICATION OF DEFECTS (Cont'd)

MINOR: AQL 4.0%	Req't Par.	QA. Par..	Test Method
201. Improper spin over of pivot	Dwg FT689472		Visual
202. Illegibility of marking	3.7	4.6.3	Visual
203. Improper identification marking	3.10	4.6.6	Visual
204. Workmanship	3.4	4.6.7	Visual

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level I, and Sampling Plan Table V of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%	Req't Par.	QA. Par.	Test Method
101. Operating temperature	3.5.1	4.6.1.1	Standard Measure Equipment(SME)
102. Storage temperature	3.5.2	4.6.1.2	SME
103. Evidence of damage or defects subsequent to storage temperature test	3.5.2	4.6.1.2	Visual-Tactile

4.6 Test methods.

4.6.1 Environmental testing.

4.6.1.1 Operating temperature.- The operating temperature test shall be conducted in accordance with Procedure I of Specification MIL-F-13926, except that the conditions specified in 3.5.1 shall apply. When thermal stabilization is reached at the specified exposures, respectively, the plotting board shall be tested in accordance with 4.6.2 through 4.6.5 to determine cornpliance with the requirements of 3.5.1.

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4..6.1.2 Storage temperature.- Employing the same equipment that is capable of maintaining the specified temperatures until thermal stabilization is reached as used for the temperature test of 4.6.1.1, the storage temperature test shall be conducted. in accordance with Procedure II of Specification MIL-F-13926. However, the temperatures specified in 3.5.2 shall apply. Upon completion of the specified exposure, at which time the plotting board shall be visually and manually inspected for evidence of damage or defects, after which the plotting boards shall be tested in accordance with 4.6.2 through 4.6.5 to determine compliance with the requirements of 3.5.2.

4.6.2 Operability.- The requirements of 3.6 for operability test shall be performed manually. The dial shall vbe removable with normal pressure by the forefinger along the periphery of the dial and the ball of the thumb on the pivot pin.

4.6.3 Marking and legibility.- The dial surface of the plotting board shall be capable of accepting clear and legible markings from a load pencil of a hardness of 2H under normal writing pressure, and upon erasure with a commerical pencil eraser those shall be no evidence of permanent marks, smudges or smears. The surface shall be free of the defects speciifed in 3.7 the degree or extent of which in, the judgment of the Government Represenative, would not be mistaken by pencil marks. Lines and figures shall be visually inspected to the requirement of 3.7 to determime Compliance therein.

4.6.4 Accuracy.- When one end of the index line of the dial is in coincoincidence with either end of the "x" or "y" axis of the base, the opposite end of the index line must coincide with The opposite and of either the "x" or "y" axis within the tolerance specified in 3.8 Axis "x" and "y" shall be considered as the longest center lines of the grid.

4.6.4.1 A check of concentricity between the mil graduations of the dial and grid spacings of the base with the respect to the pivot point shall be accomplished by laying gun anti target positions on the dial and reading gun target positions on the graduated scale as outlined. below.

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4.6.4.1.1 Coincide the 800 mil graduation (target azimuth from the observer OT) in coincidence with the zero regerence line on the base and place a pencil mark on the dial at the 2500 meter range mark above axis "x". Coincide the 2500 mil graduation (gun azimuth from the observer OG) to the zero reference line and place a pencil mark on the dial at the 1250 range marks. By rotating the dial, align the constructed pencil line parallel to axis "y" and read the gun target (azimuth GT) at the zero reference line. This azimuth reading must coincide with the gun target GT azimuth specified in 3.8.1. Repeat this procedure for the remaining problems specified in 3.8.1 placing the marks in their proper quadrants.

4.6.5 Warpage.- Warpage between the edge of the dial and the base shall be checked by measuring the distance from the top surface of the dial to the surface of the base around the perimeter of the dial by means of standard, measuring equipment. The difference in measurements is considered warp and shall not exceed the tolerance specified in 3.9.

4.6.6 Identification marking.- Marking for identification shall be visually inspected for conformance in accordance with Standard MIL-STD-130.

4.6.7 Workmanship.- Workmanship shall be in accordance with Specification MIL-F-13926.

4.6.8 Packaging, packing and marking.- The quality assurance provisions for the preservation, packaging, packing, and marking for shipment shall be in accordance with the document specified in 5.1

5. PREPARATION FOR DELIVERY

5.1 Packaging,packing, and marking.-- The plotting board shall be packaged, packed, and marked in accordance with Specification MIL-F-14619 P 13327. The level of protection shell. be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The plotting board is used for reading directly the distance of targets in meters from the gun by plotting individually the location of the target in respect to the gun as obtained at the observation post.. Direction in azimuth is also obtained in mils by this same procedure.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Selection of applicable levels of preservation, packaging, and packing
(c) Applicable stock number
(d) Provisions for production model including date of delivery (see 3.1)

6.3 Definitions.

6.3.1 Standards measuring equipment (SME).- Standard measuring equipment includes the common hand type measuring devices which are usually stocked by commerical supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commerical testing equipment such as meters, optical comparators, etc.

6.3.2 Tactile.- Skillful manipulation or sensing.

6.3.3 Production model.- A model in its final mechanical and electrical form of final production design made by production methods, tools, jigs and fixtures.

6.4 Operation of Plotting Board: M17.- Operation of the plotting board is described in Technical Manual TM9-1220-223-15, "Operator, Organizational, Field and Depot Maintenance Manual (Including Repair Parts and Special Tools List)".

NOTICE: When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responspility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, Specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

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Army-Ord

Preparing activity:

Army-Ord

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